ACAV Project – Summary of tasks T03 and T04

Path geometry, corner track and Frenet frame

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1 Role in the project

The project studies overtaking between two cars in a single corner. Every module (speed profile, racing line, planners, simulator) needs a common basis: *a track* and *a way to describe a position and a path on that track*. T03 and T04 provide this basis (file `src/overtake/track.py`). They lie on the critical path: T03 → T07 → T08 → T14 → T16 → ...

Conventions. SI units; heading ψ measured counter-clockwise from the x axis; signed curvature κ (written ρ in the course), positive for a left turn; Frenet frame (s, d) with s the arc length along the centreline and d the lateral offset, $d > 0$ to the left.

2 T03 – Geometry of a path

Goal. For any path given by an ordered sequence of points $(x_i, y_i)_{i=0..N-1}$ (centreline, racing line, overtaking candidate, lattice path), compute at each point the distance travelled, the heading and the curvature.

- `arc_length(x, y)` $\to s$: cumulative distance,
$$s_0 = 0, \qquad s_{i+1} = s_i + h_i, \qquad h_i = \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}.$$
- `heading(x, y)` $\to \psi$: at an interior point, tangent of the circle through the point and its two neighbours, obtained by weighting each segment direction $\mathbf{e}$ by the length of the other segment: $\mathbf{t}_i \propto h_i\,\mathbf{e}_{i-1} + h_{i-1}\,\mathbf{e}_i$. The result is exact on straight lines and circles, even with uneven spacing. At the end points we use the fact that the direction of a chord is the mean of the tangents at its two ends. The heading is unwrapped (no 2π jumps).
- `curvature(x, y)` $\to \kappa$ (slides 6b, *Feasibility check*): three consecutive points form a triangle with sides a, b, c and area A. The circle through these three points has radius $abc/(4A)$, hence
$$\kappa_i = \frac{4A}{abc}, \qquad 2A = \Delta x_{i-1}\,\Delta y_i - \Delta y_{i-1}\,\Delta x_i \quad \text{(signed cross product)}.$$
The sign of the area gives the direction of the turn.

Why curvature matters. In a corner, $a_y = v^2\kappa$. Since the tyres limit $a_y \le a_{y,\max}$, the curvature sets a local maximum speed
$$v_{\max}(s) = \sqrt{a_{y,\max}/|\kappa(s)|},$$
for example 15 m/s in the hairpin ($R = 15$ m, $a_{y,\max} = 15$ m/s^2) and 24.5 m/s for $R = 40$ m. It is then used in:

- **T07** (three-pass speed profile, 6a): initial guess, then sharing of the friction ellipse between $a_y = v^2\kappa$ and a_x;
- **T08** (racing line, L7): cost $J = k_l \sum \Delta s_i^2 + k_\rho \sum \kappa_i^2$;
- **T16** (sigmoid, 6a): choice of the steepest feasible sigmoid;
- **T19** (lattice, 6b): a node is feasible if $\kappa \le \rho_{\max}(v) = \min\left(\rho_{\max,\text{steer}},\ a_{y,\max}/v^2\right)$.

3 T04 – Corner track and Frenet frame

Goal. Build the corner of the proposal (Section 3.1) and convert between global coordinates (x, y) and Frenet coordinates (s, d) (slides 6a). The planners reason in (s, d); the simulator and the vehicle models work in (x, y).

`CornerTrack(l_in, radius, angle_deg, l_out, width, ds).` Entry straight of length L_{in}, arc of radius R and angle Θ (> 0 for a left turn), exit straight L_{out}, constant width w. By default, the hairpin of the proposal: $R = 15$ m, $\Theta = 180°$, $w = 12$ m. We have

$$s_{\text{start}} = L_{in}, \qquad s_{\text{end}} = L_{in} + R|\Theta|, \qquad L = s_{\text{end}} + L_{out}.$$

The centreline is sampled every ≈ 0.5 m in the arrays `s, cx, cy, psi, kappa` (the "stations" used by T07 and T08).

Methods.

- `at_s(s)` $\rightarrow (x, y, \psi, \kappa)$ of the centreline, analytically. With φ the angle travelled in the arc and $\sigma = \pm 1$ the direction of the turn:

$$x = s_{\text{before}} + R\sin\varphi + s_{\text{after}}\cos\psi, \qquad y = \sigma R(1 - \cos\varphi) + s_{\text{after}}\sin\psi,$$
$$\psi = \sigma\varphi, \qquad \kappa = \sigma/R \text{ in the arc, 0 elsewhere.}$$

- `from_frenet(s, d)` $\rightarrow (x, y)$: the centreline point is moved by d along the left normal $\hat{n} = (-\sin\psi,\ \cos\psi)$,

$$\mathbf{r}(s, d) = \mathbf{c}(s) + d\,\hat{n}(s).$$

  This is the formula $\hat{r}(s) = r(s) + y(s)\,\hat{n}(s)$ of the sigmoid planner (6a) and the placement of the waypoints "on the normal, at distance d from the check point" of the lattice planner (6b).

- `to_frenet(x, y, s_hint)` $\rightarrow (s, d)$: each point is projected on the three pieces (straight, arc, straight), the closest projection is kept, and d is the component along $\hat{n}$. In the hairpin the two straights are only $2R = 30$ m apart, so a point can be equally close to both. `s_hint` (the s of the previous time step) restricts the search to $[s_{\text{hint}} \pm 20$ m$]$ and removes the ambiguity. Used by the simulator (T11) at every step.

- `d_limits(car_width, margin)`: equation (1) of the proposal, $\alpha \in [\Delta\alpha, 1 - \Delta\alpha]$ with $\Delta\alpha = (w_v/2 + e)/w$, rewritten in d:

$$|d| \le \frac{w}{2} - \frac{w_v}{2} - e \qquad \text{(e.g. } 6 - 0.95 - 0.3 = 4.75 \text{ m)}.$$

  It bounds the racing-line optimisation (T08) and the offsets of the planners (T16, T19).

- `bounds()`: left and right edges $\mathbf{r}(s, \pm w/2)$, for plots ("inner/outer bounds" representation of L7).

Two arc lengths not to confuse. `track.s` measures the distance along the *centreline* (a coordinate shared by both cars), whereas `arc_length(x, y)` measures the distance actually travelled along a given path. A car on the inside of the corner travels fewer metres than `track.s` suggests; the speed profile must therefore use the length and curvature of the actual path.

4 Validation

Test	**Check**
Circle of radius R	$\kappa = +1/R$ turning left, $-1/R$ turning right
Straight line	$\kappa = 0$
Uneven spacing	curvature and heading still exact
Track length	$L = L_{in} + R\lvert\Theta\rvert + L_{out}$
T03/T04 consistency	`curvature(cx, cy)` $\approx$ `kappa`, same for s and ψ
Frenet round trip	`to_frenet(from_frenet(s, d))` $= (s, d)$ up to 10^{-9}
Side of d	$d > 0$ on the left, hence the inside of a left turn
Hairpin	`s_hint` picks the right straight
Limits	`d_limits` $= \pm 4.75$ m; inside edge shorter than outside edge

All 26 tests pass. The script `scripts/plot_track.py` plots the track, its edges, the limits of the car centre, the frame $(\hat{t}, \hat{n})$, and the exact $\psi(s)$ and $\kappa(s)$ compared with the values recomputed by T03.